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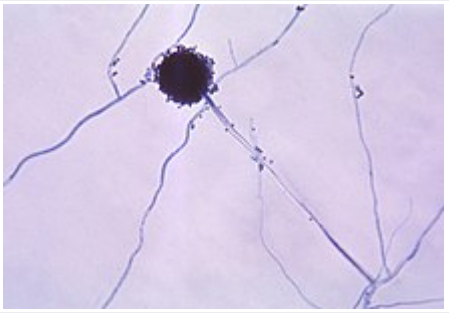
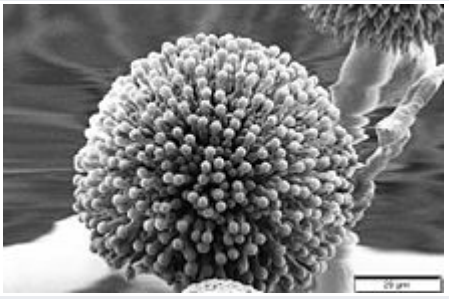
Aspergillus niger

Aspergillus niger is a mold classified within the *Nigri* section of the *Aspergillus* genus.[1] The *Aspergillus* genus consists of common molds found throughout the environment within soil and water, on vegetation, in fecal matter, on decomposing matter, and suspended in the air.[2] Species within this genus often grow quickly and can sporulate within a few days of germination.[2] A combination of characteristics unique to *A. niger* makes the microbe invaluable to the production of many acids, proteins and bioactive compounds. Characteristics including extensive metabolic diversity, high production yield, secretion capability, and the ability to conduct post-translational modifications are responsible for *A. niger's* robust production of secondary metabolites.[3] *A. niger's* capability to withstand extremely acidic conditions makes it especially important to the industrial production of citric acid.[1][4]

A. niger causes a disease known as "black mold" on certain fruits and vegetables such as grapes, apricots, onions, and peanuts, and is a common contaminant of food. It is ubiquitous in soil and is commonly found in indoor environments, where its black colonies can be confused with those of *Stachybotrys* (species of which have also been called "black mold").[5] *A. niger* is classified as Generally Recognized as Safe (GRAS) by the US Food and Drug Administration for use in food production,[6] although the microbe is capable of producing toxins that affect human health.[7]

Taxonomy

Aspergillus niger is included in *Aspergillus* subgenus *Circumdati*, section *Nigri*. The section *Nigri* includes 15 related black-spored species that may be confused with *A. niger*, including *A. tubingensis*, *A. foetidus*, *A. carbonarius*, and *A. awamori*.[8][9] In 2004, a number of morphologically similar species were described by Samson *et al.*[9]

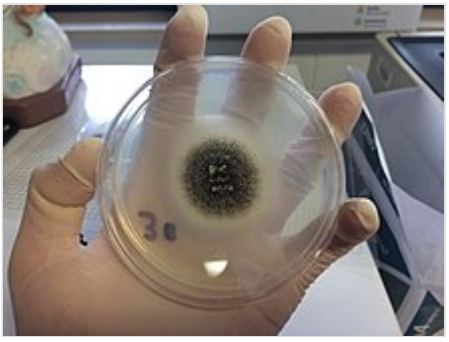
Aspergillus niger	
	
Photomicrograph showing the conidial head (conidiophore) of *Aspergillus niger*	
	
Details of the head by scanning electron microscopy	
Scientific classification	
Domain:	Eukaryota
Kingdom:	Fungi
Division:	Ascomycota
Class:	Eurotiomycetes
Order:	Eurotiales
Family:	Aspergillaceae
Genus:	*Aspergillus*
Species:	***A. niger***
Binomial name	
Aspergillus niger van Tieghem 1867	
Synonyms	

In 2007, the strain of ATCC 16404 *Aspergillus niger* was reclassified as *Aspergillus brasiliensis* (refer to publication by Varga et al.[10]). This required an update to the U.S. Pharmacopoeia and the European Pharmacopoeia, which commonly use this strain throughout the pharmaceutical industry.

Aspergillus niger var. niger
Aspergillopsis nigra (Tiegh.) Speg.
Rhopalocystis nigra (Tiegh.) Grove
Sterigmatocystis nigra (Tiegh.) Sacc. (1877)

Cultivation

A. niger is a strict aerobe; therefore, it requires oxygen to grow.[11] The fungus can grow in a range of environmental conditions; it can grow at temperatures ranging from 6 to 47 °C.[12] As a mesophile,[13] its optimal temperature range is 35-37 °C.[11] It can tolerate pH ranging from 1.5 to 9.8.[12] *A. niger* is xerophilic, meaning it can grow and reproduce in environments with very little water. It can also grow in humid conditions even tolerating environments with 90-100% relative humidity.[13] The fungus is most commonly grown on potato dextrose agar (PDA), but it can grow on many different types of growth media including Czapek-Dox agar, lignocellulose agar, and several others.



A. niger growing on potato dextrose agar

Genome

Aspergillus niger has a genome consisting of roughly 34 megabases (Mb) organized into eight chromosomes.[14] The DNA contains 10,785 genes which are transcribed and translated into 10,593 proteins.[14]

Two strains of *A. niger* have been sequenced. Strain CBS 513.88 produces enzymes used in industrial applications while strain ATCC 1015 is the wildtype strain of ATCC 11414 used to produce industrial citric acid (CA).[15][16][17] The *A. niger* ATCC 1015 genome was sequenced by the Joint Genome Institute in a collaboration with other institutions.[18] Completed sequences have been used to uncover orthologous genes and pathways involved in fungal metabolism, specifically the catabolism of monosaccharides.[19] The ability of *A. niger* to change its metabolism depending on the carbon sources and other nutrients present in its environment has enabled the microorganism to survive and be found in almost all ecosystems. Further research is being done to study these mechanisms for all fungi using the complete sequenced genome of *A. niger*.[19]

Genomic information

NCBI genome ID	429 (https://www.ncbi.nlm.nih.gov/genome/?term=429)
Ploidy	haploid
Genome size	34 Mb
Number of chromosomes	8

Industrial uses

There are two ways in which *Aspergillus niger* can be grown for industrial purposes: solid state fermentation (SSF) and submerged fermentation (SmF).[20] SSF uses a solid substrate with nutrients and minimal moisture to grow microorganisms. Nutrients such as nitrogen and carbon come from agricultural byproducts such as wheat bran, sugar pulp, rice husks, and corn flour.[21] SSF gives better yield of microbe products and is more cost effective than SmF due to using agricultural byproducts.[22] SSF is predominantly

used over SmF.[22] In SmF, microbes are grown in a liquid medium inside large aseptic fermentation vessels.[21][22] These vessels are expensive pieces of equipment that provide more water for growth and allow for tight control of environmental factors, such as temperature and pH, that affects microbial growth.[22]

Aspergillus niger is cultured to facilitate the industrial production of many substances.[23] Various strains of *A. niger* are used in the industrial preparation of citric acid (E330) and gluconic acid (E574); therefore, they have been deemed acceptable for daily intake by the World Health Organization.[24] *A. niger* fermentation is "generally recognized as safe" (GRAS) by the United States Food and Drug Administration under the Federal Food, Drug, and Cosmetic Act.[25] *A. niger* is also being considered as a potential new source of natural food grade pigments.[26]

The production of citric acid (CA) is achieved by growing strains of *A. niger* in a nutrient rich medium that includes high concentrations of sugar and mineral salts and an acidic pH of 2.5-3.5.[27] Many microorganisms produce CA, but *Aspergillus niger* produces more than 1 million metric tons of CA annually via a fungal fermentation process.[28] CA is in high demand for applications such as the control of microorganism growth, food and beverage flavor enhancement, acidity manipulation, pharmaceuticals, etc.[29]

A. niger produces many useful enzymes for the catabolism of biopolymers in order to obtain nutrients from its environment.[30] The production of specific enzymes can be increased for industrial purposes.[31][30] For example, *A. niger* glucoamylase (P69328 (https://www.ebi.ac.uk/pdbe/pdbe-kb/proteins/P69328)) is used in the production of high-fructose corn syrup and pectinases (GH28) are used in cider and wine clarification. Alpha-galactosidase (GH27), an enzyme that breaks down certain complex sugars, is a component of Beano and several other products that decrease flatulence.[32] Another use for *A. niger* within the biotechnology industry is in the production of magnetic isotope-containing variants of biological macromolecules for NMR analysis.[33] *Aspergillus niger* is also cultured for the extraction of the enzyme, glucose oxidase (P13006 (https://www.ebi.ac.uk/pdbe/pdbe-kb/proteins/P13006)), used in the design of glucose biosensors, due to its high affinity for β-D-glucose.[34][35]

In the food industry, *A. niger* is also cultured to isolate the enzyme fructosyltransferase to produce fructooligosaccharides (FOS).[36] FOS are used to manufacture low-calorie and functional foods due to FOS characteristic ability to slow growth of pathogenic microorganisms in the intestines.[36][37] These foods have prebiotic fiber among other health promoting properties. *A. niger* is not the only organism to produce the enzyme fructosyltransferase, but it has been found to produce the enzyme at rates conducive to industrial production.[36][37] A specific use of A. niger within the food industry is its capability to produce enzymes like carbohydrase and cellulase, which are commonly used in the seafood industry for removing the bellies of clams during processing and removing the tough external skin of shrimp from their edible internal tissue.[38]

Aspergillus niger can grow in gold-mining solutions containing cyano-metal complexes with gold, silver, copper, iron, and zinc. The fungus also plays a role in the solubilization of heavy-metal sulfides.[39] *A. niger* has also been shown to remediate acid mine drainage through biosorption of copper and manganese.[40]

Toxicity

A. niger produces a wide variety of secondary metabolites,[7] some of which are mycotoxins called ochratoxins,[41] such as ochratoxin A.[5][42] Contamination by filamentous fungi, such as *A. niger,* occurs frequently in grapes and grape based products resulting in contamination by ochratoxin A (OTA). OTA, a clinically relevant mycotoxin, can accumulate in human tissue and cause a variety of serious health conditions.[43] Potential consequences of OTA poisoning include kidney damage, kidney failure and cancer but the United States Food and Drug Administration (FDA) has not set maximum permissible levels of OTA in food unlike the EU that set maximum permissible levels in a variety of food products.[44]

Pathogenicity

A. niger growing on onion

Plant pathogen

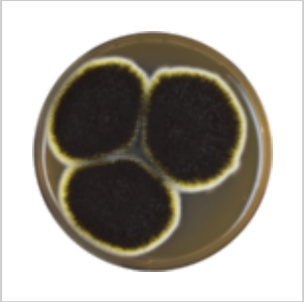
Aspergillus niger can cause black mold infections in certain legumes, fruits, and vegetables such as peanuts, grapes, and onions, leading to the fungus being a common food contaminant. This filamentous ascomycete has a tolerance to changes in pH, humidity, and heat, thriving in a temperature range from 15 to 53 °C (59 to 127 °F).[45] These characteristics make infections of *A. niger* a common cause of post-harvest decay in fruits and vegetables, which can lead to significant economic loss in the food industry.[46] *A. niger* infection in plants can cause a reduction in seed germination, seedling emergence, root elongation, and shoot elongation, causing the plant to perish before maturation.[46] Specifically, *Aspergillus niger* causes sooty mold on onions and ornamental plants.

Human pathogen

A. niger is pathogenic. Aspergillosis is a fungal infection caused by spores of indoor and outdoor *Aspergillus* mold species.[47] Due to the ubiquitous nature of *A. niger*, its spores are commonly inhaled by humans from their surrounding environment.[48] Aspergillosis infection customarily occurs in people with compromised immune systems or pre-existing lung conditions like asthma and cystic fibrosis.[47] Types of aspergillosis include allergic bronchopulmonary aspergillosis (ABPA), allergic aspergillus sinusitis, azole-resistant aspergillus fumigatus, cutaneous (skin) aspergillosis, and chronic pulmonary aspergillosis.[47] Out of the approximated 180 species of *aspergillus* molds, roughly 40 species have been found to cause health concern in immunocompromised humans.[47] Aspergillosis is particularly frequent among horticultural workers who often inhale peat dust, which can be rich in *Aspergillus niger* spores. The fungus has also been found in ancient Egyptian mummies and can be inhaled when they are disturbed.[49] Otomycosis, which is a superficial fungal infection of the ear canal, is another disorder that can be caused by overgrowth of *Aspergillus* molds like *A. niger.*[50] Otomycosis caused by *A. niger* is frequently associated with

mechanical damage of the ear canal's external skin barrier and often presents itself in patients living in tropical climates.[50][42] *A. niger* is rarely reported to cause pneumonia compared to other *Aspergillus* species, such as *Aspergillus flavus*, *Aspergillus fumigatus*, and *Aspergillus terreus*.[51]

Gallery



A. niger growing in a Petri dish

See also

- Contamination control

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External links

- *Aspergillosis* information, Centers for Disease Control and Prevention, US Department of Health and Human Services (https://www.cdc.gov/fungal/diseases/aspergillosis/index.html)
- *A. niger* ATCC 1015 genome (http://genome.jgi.doe.gov/Aspni7/Aspni7.home.html)
- *Aspergillus* website (Manchester University, UK) (http://www.aspergillus.org.uk/)

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